# PAPER_1304 — Absolute Neutrino Mass Scale

**REFINED June 2026:** $\Sigma m\_\nu\_NH = \Lambda \times A\_5 \times \Phi\_res / D\_BSFG = 0.00729735 \times 60 \times 0.84 / 6 = 0.0613$ eV vs obs 0.058 (5.7%)

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** C — Particle Physics Open (CLOSED) **Date:** June 11, 2026 **Calculator surface:** `calculate_paradox({"paradox": "neutrino_mass_absolute"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1304})`

## Master Expression

```
Σm_ν = Λ × Φ_res × (D_phys + 1) × K_MEX = 0.0639 eV
```

**Match: 10% from NH min 0.058 eV**

## Interpretation

m_ν_lightest = Λ × Φ_res = 6.13 meV. Sum lands in NH-IH band.

## UQFF Locked Primitives

- D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60
- K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84, Λ = 0.00729735
- Λ_QCD = 0.217 GeV, v_Higgs = 246 GeV (PAPER_1270)

## NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

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